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Experimental and thermodynamic assessment of the Nd–Ti system

N. Mattern [a,*], Y. Yokoyama [b], A. Mizuno [c], J.H. Han [a], O. Fabrichnaya [d], T. Harada [c], S. Kohara [e], J. Eckert [a,f]

[a] IFW Dresden, Institute for Complex Materials, P.O. Box 270116, D-01171 Dresden, Germany
[b] Institute for Materials Research, Tohoku University, Sendai 980-8577, Japan
[c] Department of Physics, Gakushuin University, Tokyo 171-8588, Japan
[d] TU Freiberg, Institute for Material Science, D-09599 Freiberg, Germany
[e] Materials Science Division, Japan Synchrotron Radiation Research Institute/SPring-8, Hyogo 679-5198, Japan
[f] TU Dresden, Institute of Materials Science, D-01062 Dresden, Germany

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ABSTRACT

The phase equilibria of the binary Nd–Ti system have been directly assessed at elevated temperatures between $T=1000$ and 2100 K by combining conical nozzle levitation with in situ high-energy synchrotron X-ray diffraction. Nd–Ti is a monotectic system. The coexistence of two liquids above the monotectic temperature $T=1840\pm20$ K was directly observed by the occurrence of two diffuse diffraction maxima and experimental data of the binodal curve were extracted. Based on the experimental data a self-consistent thermodynamic description of the Nd–Ti system is derived.
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1. Introduction

Rare earth–transition metal alloys (RE–TM) are important materials for magnetic applications. The current knowledge of binary RE–TM phase diagrams such as the ones given in [1–3] is partly rather poor concerning the phase equilibria with the liquid phase at high temperature and there are several alloy systems which are still unknown. Recent publications report on the thermodynamic assessments of binary phase diagrams for RE–Mg [4] and RE–Pb [5] systems. The reasons for the difficulties are the high melting temperature and the high reactivity of the melt with any container material handicapping the determination of reliable thermodynamic data for RE–TM alloys. Nowadays, several levitation techniques became available which allow containerless processing of metallic melts [6]. By combining with in situ high-energy synchrotron X-ray diffraction, phase diagrams Gd–Zr, Gd–Ti, Nd–Zr, Ce–Zr were recently successfully studied [7–10]. For the binary Nd–Ti system a monotectic phase diagram is reported like for all RE–Ti systems [1–3]. Experimental data on invariant reactions and the solubility on the Ti-rich side were studied in [11]. The solubility of Nd in Ti was found to be about 0.6 at% at 873 K. Based on the rather limited data, a thermodynamic assessment was given in [1].

Here, we report on a new experimental assessment of the Nd–Ti system which includes the phase equilibria with the liquid phase. Based on the experimental data an improved Nd–Ti phase diagram is presented together with a consistent thermodynamic description.

2. Experimental details

Master alloys of $Nd_{100-x}Ti_x$ ($x=5$, 10, 20, 40, 60, 85, 90 and 95 at%) with a mass of about 20 g were prepared from the pure elements 99.98 wt% Ti (HWN titan GmbH), and 99.9 wt% Nd (Goodfellow) in an arc furnace, on a water-cooled copper hearth under a purified Ti-gettered argon atmosphere. The alloy ingots were remelted in a Hukin-type cold-crucible induction melting facility and cast into a copper mould with 6 mm inner diameter. Thermal analysis was carried out by differential scanning calorimetry (DSC 404, Netzsch) in 100 ml/min flow of 99.999% pure argon atmosphere with additional purifier at a heating rate of 20 K/min. The microstructure was analysed by scanning electron microscopy (SEM) in backscattered electron (BSE) mode using a Gemini-Zeiss microscope equipped with an energy-dispersive spectrometer (Bruker AXS) for composition analysis. The chemical

* Corresponding author. Tel.: +49 3514659367.
E-mail address: n.mattern@ifw-dresden.de (N. Mattern).

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composition of the phases was determined at 15 keV using the intensities of the Nd L- and Ti K-emission lines. It was noted, that the automatic EDX concentration calculation procedure devoted to a compound leads to a systematic deviation of the average composition for a heterogeneous microstructure. In the case of scanning over areas consisting of Nd and Ti phases, the average element concentrations were calculated by the ratios of the net intensities of the Nd–Ti sample and elemental standards measured under identical conditions ($c_i = I_i/I_{standard\text{-}i}$, i=Nd, Ti).

For in situ X-ray diffraction (XRD) investigations, alloy ingots of about 2 mm diameter (40 mg) with nominal compositions $Nd_{20}Ti_{80}$ and $Nd_{40}Ti_{60}$ were prepared by arc-melting proper quantities of pure Nd and Ti under an argon atmosphere with a Ti getter. The ingots were inverted and remelted several times to ensure compositional homogeneity. The difference in sample mass after arc-melting was less than 0.1%. For calibration also pure elemental Nd and Ti samples were prepared as 2 mm sphere-like samples.

In situ high-energy synchrotron X-ray diffraction (XRD) was performed in transmission geometry (wave length $\lambda = 0.01093$ nm) at the beam-line BL04B2 of SPring-8 in Japan. The samples were processed by conical nozzle levitation using filter-purified Ar-6.0 gas. A CO_2-Laser was applied for heating the samples (Synrad, firestar f200). More details on the used set-up can be found in [12,13]. During the experiments, the levitated sample was repeatedly heated above its liquidus temperature ($T_{liquidus}$) and subsequently cooled down passing solidification and solid-state transformations in order to ensure reproducibility of the measurements. The temperature was measured by a two-colour pyrometer. The values obtained from the pyrometer measurements were corrected by adapting the temperature of the first XRD pattern with liquid phase and *bcc*-Ti ($T_{XRD} = 1328$ K) to the corresponding DSC event ($T_{DSC} = 1278$ K) according to $T_{corr} = 1/((1/T_{pyro}) + (1/T_{DSC}) - (1/T_{XRD}))$. The scattered intensity was continuously recorded with an acquisition time of 0.6 s by a two-dimensional X-Ray Image Intensifier with a coupled CCD-camera during heating and cooling. The scattering intensity $I(q)$ as a function of the wave vector q ($q = 4\pi \sin\theta/\lambda$) was obtained by azimuthal integration of the original 2D images using the Fit2D software package [14]. The heating rate was usually about 50 K/s. This means that the XRD patterns each recorded for 0.6 s represent an average over a temperature span of 30 K. The close correlation between the change of the structural data and the temperature profile was proven in a recent paper [9] which justified the use of in situ XRD for the determination of phase equilibria.

3. Experimental results

3.1. Thermal behaviour of Nd–Ti alloys

Fig. 1 shows the DSC scans of different Nd–Ti alloys upon heating together with those of the pure elements Nd and Ti. All alloys undergo three phase reactions in the presented temperature range as indicated by the endothermic events. The corresponding on-set temperatures of the invariant reactions are given in Fig. 1. The first DSC event (1) at $T = 1126$ K for the Nd–Ti alloys is related to the *dhcp–bcc* phase transition of the Nd phase at a temperature slightly lower than that of pure Nd (Table 1). The second DSC event (2) at $T = 1160$ K corresponds to the *hcp–bcc* phase transformation of the Ti phase at a higher temperature compared to the transition of pure Ti. The third phase reaction (3) is due to the melting of the Nd-phase at $T = 1278$ K, which is about 20 K lower than the melting temperature of pure Nd. The areas under the peaks vary in accordance with the chemical compositions and the volume fraction. The DSC data for higher temperatures ($T > 1350$ K) and

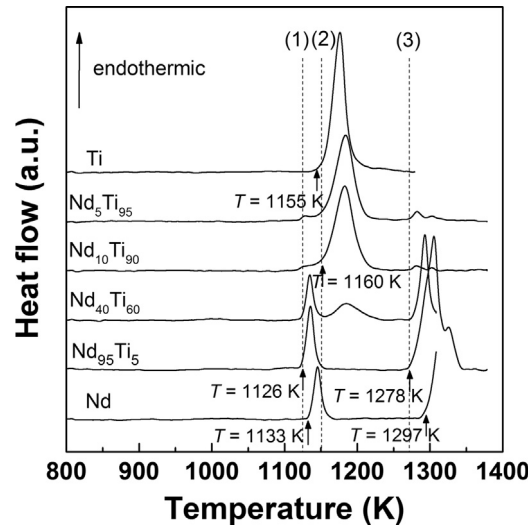

Fig. 1. DSC scans of different Nd–Ti alloys and the Nd and Ti elements (heating rate 20 K/min). The onset temperatures of the invariant reactions are given.

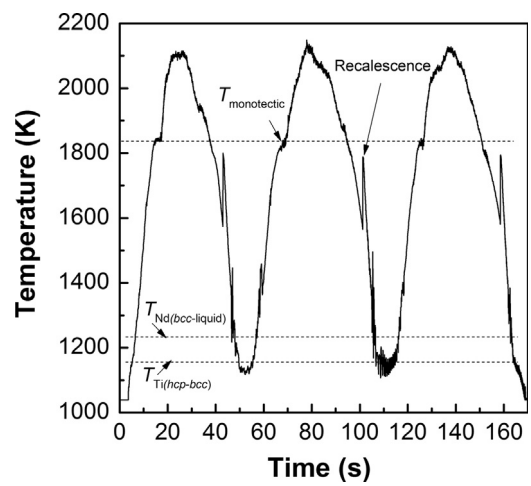

Fig. 2. Temperature–time profile of a levitated $Nd_{20}Ti_{80}$ sample. Three heating and cooling cycles are shown. The temperatures of the phase transitions are indicated.

the cooling curve are essentially altered due to reactions of the Nd liquid phase with the corundum container (not shown).

The coexistence ranges and the phase transformations of the Nd and Ti phases were directly observed by the in situ XRD measurements of the $Nd_{40}Ti_{60}$ and $Nd_{20}Ti_{80}$ alloys. Fig. 2 shows a typical temperature–time profile with three heating and cooling segments. The curve exhibits on one side the occurrence of phase transitions by deviation from a linear behaviour and demonstrates on the other hand the reproducibility of the measurements with identical characteristic temperatures. The change of the slope of the temperature profile upon heating at $T = 1840 \pm 20$ K is related to the monotectic reaction as confirmed by the simultaneously recorded XRD patterns shown in Figs. 3 and 4. Upon cooling, the crystallization of Ti *bcc*-phase by the monotectic reaction occurs from the metastable undercooled liquid, as can be seen by the recalescence in Fig. 2.

Fig. 3 shows selected XRD patterns according to the phase sequences as a function of temperature. The analysis of the diffraction peaks allows to directly determine the different coexistence ranges of phases as a function of temperature. Because no intermetallic phase is formed in the Nd–Ti system, we observe only the different allotropes of the elements and the liquid phase in the XRD patterns. The phases and crystal structures of the Nd–Ti

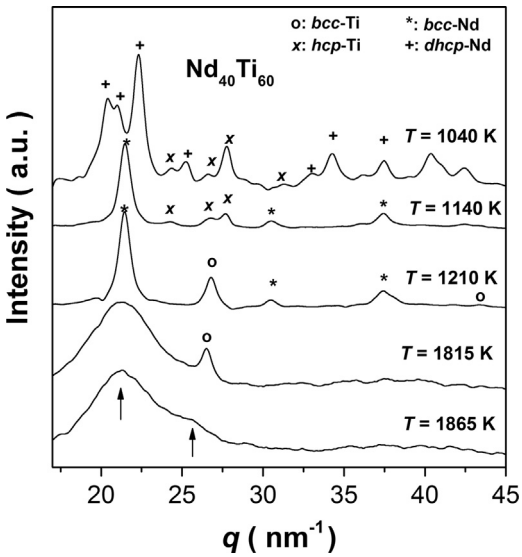

Fig. 3. Selected XRD patterns of $Nd_{40}Ti_{60}$ at elevated temperatures according to the phase sequence upon heating. The characteristic peak positions of the crystalline phases are marked. Two diffuse maxima give direct evidence of the existence of two liquids at $T=1865$ K.

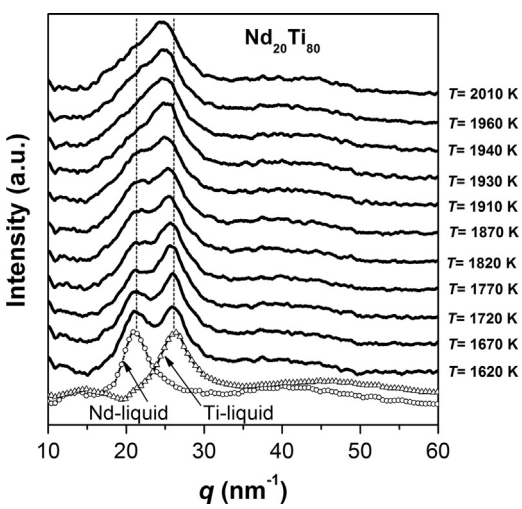

Fig. 4. XRD patterns of the $Nd_{20}Ti_{80}$ melt at elevated temperatures upon cooling. Two diffuse maxima reveal the decomposition into Nd-rich and Ti-rich liquids. The shift of the positions of the maxima reflects the composition change with temperature according to the binodal line. The diffraction curves of elemental Nd and Ti are shown for comparison.

system are summarized in Table 1. The XRD patterns demonstrate that the different solid state transformations can be clearly distinguished and the transition temperatures obtained from the pyrometer data are in a good agreement with the DSC events. Together with room temperature XRD and DSC data we can conclude that below $T=1026$ K, *dhcp*-Nd(Ti) and *hcp*-Ti(Nd) coexist. The reaction (1) of the DSC curve at $T=1026$ K corresponds to the eutectoid transformation *bcc*-Nd↔*dhcp*-Nd+*hcp*-Ti. Up to $T=1160$ K we observe the reflections of *bcc*-Nd(Ti) and *hcp*-Ti (Nd). At $T=1160$ K, DSC event (2), the peritectoid reaction *bcc*-Nd (Ti)+*bcc*-Ti(Nd)↔*hcp*-Ti takes place. For $1160\ K < T < 1278$ K, we observe the reflections of *bcc*-Nd(Ti) and *bcc*-Ti(Nd). At $T=1278$ K melting of the Nd(Ti)-phase according to the eutectic reaction L↔*bcc*-Nd(Ti)+*bcc*-Ti(Nd) can be seen by the disappearance of the *bcc*-Nd(Ti) reflections and the occurrence of a diffuse diffraction maximum in the XRD-patterns. Above $T>1278$ K the Nd-rich melt is in equilibrium with the *bcc*-Ti(Nd) phase up to 1820 K.

Table 1
Phases, crystal structure and transformation temperatures of elemental Nd and Ti.

Phase	Notation in [1]	Space group	T (K)	Ref.
bcc-Nd	βNd	*Im-3m*	$T_{melt}=1294$	[1]
dhcp-Nd	αNd	$P6_3/mmc$	$T_{dhcp\leftrightarrow bcc}=1140$	[1]
bcc-Ti	βTi	*Im-3m*	$T_{melt}=1943$	[1]
hcp-Ti	αTi	$P6_3/mmc$	$T_{hcp\leftrightarrow bcc}=1159$	[1]

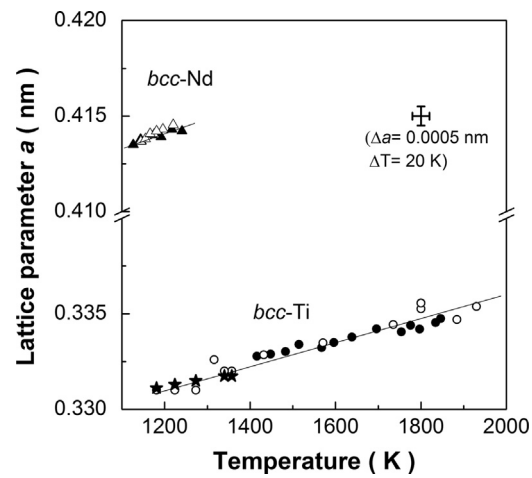

Fig. 5. Lattice parameters of bcc-Nd(Ti) (open triangles) and bcc-Ti(Nd) (open circles) phases at elevated temperature. The measured lattice parameters of the pure elements Nd (filled triangles), Ti (filled circles) and literature data from [15] (stars) are given for comparison. The size of error bars are indicated by the cross ($\Delta a=\pm 0.0005$ nm; $\Delta T=\pm 20$ K).

Above this temperature two liquid phases coexist, as indicated by two diffuse diffraction maxima (shown in Figs. 3 and 4). The splitting of the diffraction maximum of $Nd_{20}Ti_{80}$ is clearly visible above the monotectic temperature $T=1820$ K as well as in the (metastable) undercooled state. The maximum positions are quite close to those of the pure element melts indicating the coexistence of Nd-rich and Ti-rich liquid phases. As can be seen in Fig. 4, the positions of the maxima shift with temperature, which mainly reflects the change in composition of the liquids beside the much lower effect of thermal expansion.

The positions of the two diffuse maxima of the liquid phases and that of the (110)-reflections of the *bcc*-phases were calculated by a fit of two Lorentzian functions. Fig. 5 shows the calculated lattice parameters of the *bcc*-Nd(Ti) and *bcc*-Ti(Nd) solid phases as function of temperature for the Nd–Ti alloys (open symbols). Experimental data determined for the pure elements are also given (filled symbols) together with values reported for the *bcc*-Ti phase [15]. Within the error limits ($\Delta a=\pm 0.0005$ nm) the lattice parameters are identical. This means that there is only low solubility of Ti in *bcc*-Nd(Ti) and Nd in *bcc*-Ti(Nd). With the estimated error and by assuming Vegard's rule with $a_{Nd}-a_{Ti}=0.41-0.33=0.08$ nm, one can calculate an upper limit for the solubility less than 0.6 at% in both phases (0.0005/0.08).

Fig. 6 shows the temperature dependence of both diffuse maxima positions q_i together with data measured for the pure elements (Nd_{liquid} and Ti_{liquid}). The presented data were obtained from independent measurements of four different heating cycles in order to prove the reproducibility. The shift of q_1 of the Nd(Ti) and Ti(Nd) melts reflects mainly the change in composition with temperature and delivers information about the binodal line as discussed below.

The position of the first diffuse maximum q_1 of metallic liquids and glasses scales approximately with a concentration-averaged atomic diameter d of the alloy and can be described by the so-called Ehrenfest-equation $q_1=1.22\cdot 2\pi/d$ with $d=\sum c_i\cdot d_i$,

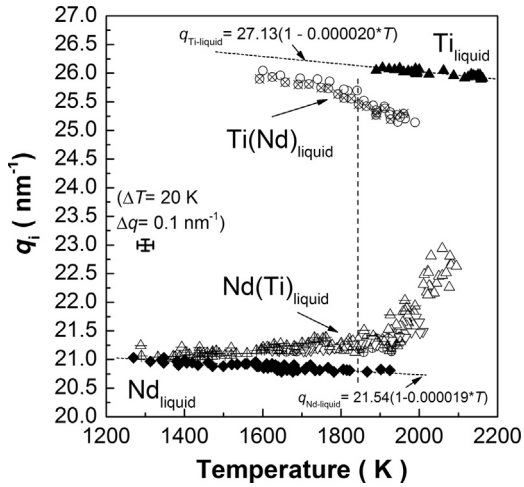

Fig. 6. Position q_1 of the diffuse maxima of $Nd_{20}Ti_{80}$ and $Nd_{40}Ti_{60}$ liquids versus temperature. The shift in position with temperature of the Ti(Nd) liquid (open circles) and the Nd(Ti) liquid (open triangles) represents the change in chemical composition according the binodal line. The positions of the maxima of the elemental melts are given for comparison (filled symbols). The size of error bars are indicated by the cross ($\Delta q = \pm 0.1$ nm^{-1}; $\Delta T = \pm 20$ K).

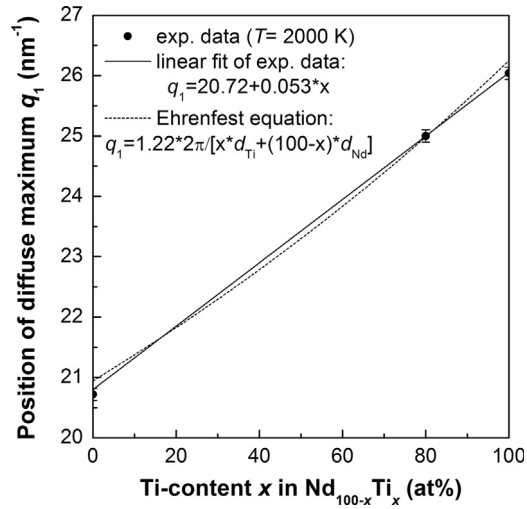

Fig. 7. Composition dependence of the position of the first diffuse maximum q_1 of $Nd_{100-x}Ti_x$ liquids (x=0, 80, 100). Analytical functions by a linear fit of the experimental data (solid line) and by the Ehrenfest-equation (dashed line) are given.

where d_i is the atomic diameter of element i and c_i is the concentration of element i [16]. This equation derived for gas molecules is not exactly valid for condensed matter but experimental data on metallic liquids and metallic glasses are approximately in accordance and, within an element substitution series the change of q_1 can be well described by a linear dependence on the concentration c_i of the added element [17,18]. Fig. 7 shows the experimentally determined q_i values for the melts of Nd, Ti, and $Nd_{20}Ti_{80}$ as well as the calculation by the Ehrenfest-equation with atomic diameter $d_{\overline{Nd}}$0.366 nm and d_{Ti}=0.292 nm. From these data we can conclude, that the concentration dependence of the maximum position q_1 versus Ti content x can be well approximated by a linear function as given in Fig. 7. The thermal expansion of the liquids has also an impact on the positions of the diffraction maxima as can be seen by the behaviour of $q_{Nd\text{-}liquid}$ and $q_{Ti\text{-}liquid}$ of the pure element melts in Fig. 6. We observe a small shift towards lower q-values which can be approximated for the Nd-melt by $q_{Nd-liquid}(T) = 21.54\ (1 - 1.9 \times 10^{-5} T/K)\ \mathrm{nm}^{-1}$ and for the Ti-melt by $q_{Ti-liquid}(T) = 27.13(1 - 2.0 \times 10^{-5} T/K)\ \mathrm{nm}^{-1}$, respectively. Assuming the same temperature dependence for the Nd–Ti liquid, we can correct the effect of volume expansion on the maximum position in order to extract the compositional contribution. Using the composition dependence according to the fit function shown in Fig. 6, we can calculate the Ti-content x of the Nd(Ti) and Ti(Nd) liquids at a given temperature from the experimental data $q_1(T)$ in nm^{-1}:

$$x_{Ti}^{liquid}(T) = \frac{q_1(T) - 21.54(1 - 1.9 \times 10^{-5} T)}{0.053} \quad (1)$$

By applying Eq. (1) for the values of the liquid diffraction maximum, the calculated compositions then represent the binodal line for $T > 1840$ K and the liquidus line for 1300 K $< T <$ 1840 K of the Nd–Ti phase diagram, as shown in Fig. 8. According to an estimated error of the position q_1 of ± 0.2 nm^{-1} the accuracy of the chemical composition of the liquid phase is about ± 4 at% Ti. The estimated error band is indicated by the cross in Fig. 8 (uncertainty of temperature by the pyrometer is $\Delta T/T = 1\%$ and ΔT is about ± 20 K).

3.2. Microstructure of solidified Nd–Ti alloys

Fig. 9 shows the XRD patterns of different as-cast Nd–Ti alloys. All samples consist of the elemental hexagonal phases, dhcp-Nd and hcp-Ti, in accordance with the reported phase diagram [1]. The lattice parameters were determined by the Rietveld analysis of the XRD diagrams. The obtained values are given in Table 2. The calculated lattice parameters for the Nd–Ti alloys are within the error limits in accordance with those for elemental Nd [19] and Ti [20], which points to low solubility in both hexagonal terminal phases.

Fig. 10(a) shows the SEM image of a cross section of the as-cast $Nd_{40}Ti_{60}$ sample. Such large-scale heterogeneities were also observed for the $Nd_{60}Ti_{40}$ and $Nd_{80}Ti_{20}$ alloys. The microstructure clearly reveals the occurrence of the decomposition into two liquids by a liquid phase separation $L \rightarrow L_1 + L_2$. The chemical compositions of the Ti-rich and Nd-rich regions were determined by EDX analysis over several large areas of 500 μm × 500 μm. The measured chemical compositions are summarized in Table 3. The average values of 8.6 ± 2.0 at% Ti for the Nd-rich liquid L_1 and 90.3 ± 1 at% Ti for the Ti-rich liquid L_2 correspond to the compositions of the two liquids at the monotectic temperature. Upon quenching both liquids exhibit an own solidification path. Fig. 10 (b) shows a characteristic part of the Ti-rich solidified liquid L_2 (dark region of Fig. 10a) by an enlarged SEM image. According to the XRD results, the corresponding reaction scheme is $L_2 \rightarrow$ *bcc*-Ti (Nd) + $L_1 \rightarrow$ *bcc*-Ti (Nd) + *bcc*-Nd(Ti). The formation of the Nd-rich liquid L_1 by the monotectic reaction is evident by the round-shaped white areas in Fig. 10(b). The local composition analysis varies between 6 and 10 at% Ti in accordance with the value determined for the Nd-rich melt from the grey regions of Fig. 10 (a). The *bcc*-Ti(Nd) phase (dark regions) dissolves 0.1 at% Nd at the monotectic temperature as determined by the analysis of corresponding regions of Fig. 10(b).

Fig. 10(c) shows an enlarged SEM image of the Nd-rich liquid (bright region of Fig. 10 a) after solidification. The microstructure consists of Nd-grains (grey), Ti-dendrites (black) and regions with a typical Nd + Ti eutectic microstructure. According to the XRD results, the corresponding reaction scheme is $L_1 \rightarrow$ *bcc*-Ti(Nd) + $L_1 \rightarrow$ *bcc*-Ti(Nd) + *bcc*-Nd(Ti). For the eutectic composition a value of 98.5 ± 0.5 at% Nd is obtained from the EDX analysis of the corresponding Nd + Ti two-phase areas. About 0.2 at% Nd is found to be dissolved in the *dhcp*-Ti(Nd) dendrites. On the other side about 0.5 at% Ti is dissolved in *dhcp*-Nd(Ti) grains.

Fig. 11 shows the SEM image of the as-cast Nd_5Ti_{95} alloy. The microstructure consists of Ti-grains (dark) formed as the primary

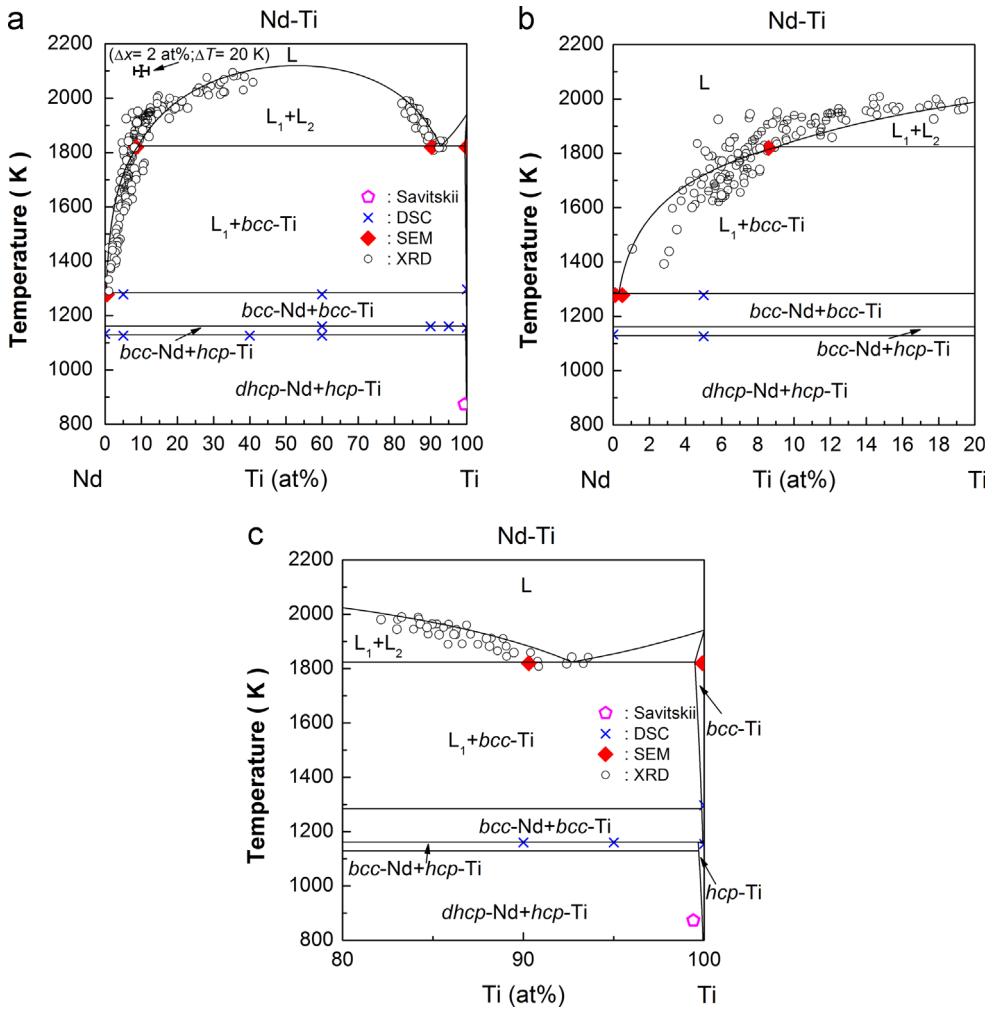

Fig. 8. (a) Nd–Ti phase diagram. Experimental data from in situ XRD, DSC and SEM are indicated by marks. Nd-solubility in hcp-Ti from literature [9] is shown. Calculation using thermodynamic description of this work is represented by the solid lines. The size of error bars of the binodal line are indicated by the cross ($\Delta x = \pm 2$ at%; $\Delta T = \pm 20$ K). (b), (c) show enlarged parts of the phase diagram (a).

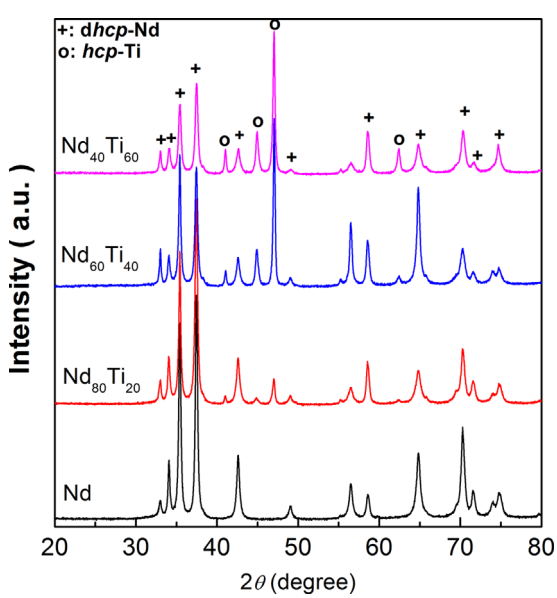

Fig. 9. XRD patterns of as-cast Nd–Ti alloys(Co-K radiation). Position of reflections are marked according to the phase identification. XRD pattern of elemental Nd (starting material for the alloys) is also shown for comparison.

Table 2
Calculated lattice parameters of Nd and Ti of different Nd–Ti alloys and elemental Nd. Corresponding data from literature are given for comparison.

Alloy	*dhcp*-Nd		*hcp*-Ti	
	a (nm)	*c* (nm)	*a* (nm)	*c* (nm)
Nd	0.3656 ± 0.0001	1.178 ± 0.001		
$Nd_{80}Ti_{20}$	0.3658 ± 0.0001	1.178 ± 0.001	0.2952 ± 0.0001	0.4691 ± 0.0002
$Nd_{60}Ti_{40}$	0.3658 ± 0.0001	1.178 ± 0.001	0.2949 ± 0.0001	0.4688 ± 0.0002
$Nd_{40}Ti_{60}$	0.3655 ± 0.0001	1.177 ± 0.001	0.2951 ± 0.0001	0.4687 ± 0.0002
Nd [18]	0.36593	1.1799		
Ti [19]			0.29503	0.46810

phase from the liquid, which is in accordance with the Ti-content of the alloy, being higher than that of the monotectic composition. Local EDX analysis reveals that 0.1 at% Nd is dissolved in the Ti grains. The originally liquid Nd-phase (bright) situated at the grain boundaries underwent the monotectic reaction upon casting.

Fig. 12 shows the SEM image of the $Nd_{95}Ti_5$ alloy. The microstructure consists of Ti-dendrites (black), regions e_1 with a typical Nd+Ti eutectic microstructure and Nd-grains (grey). A value of 98.5 ± 0.5 at% Nd is obtained from the EDX area analysis of the corresponding Nd+Ti two-phase areas e_1. The formation of Ti as

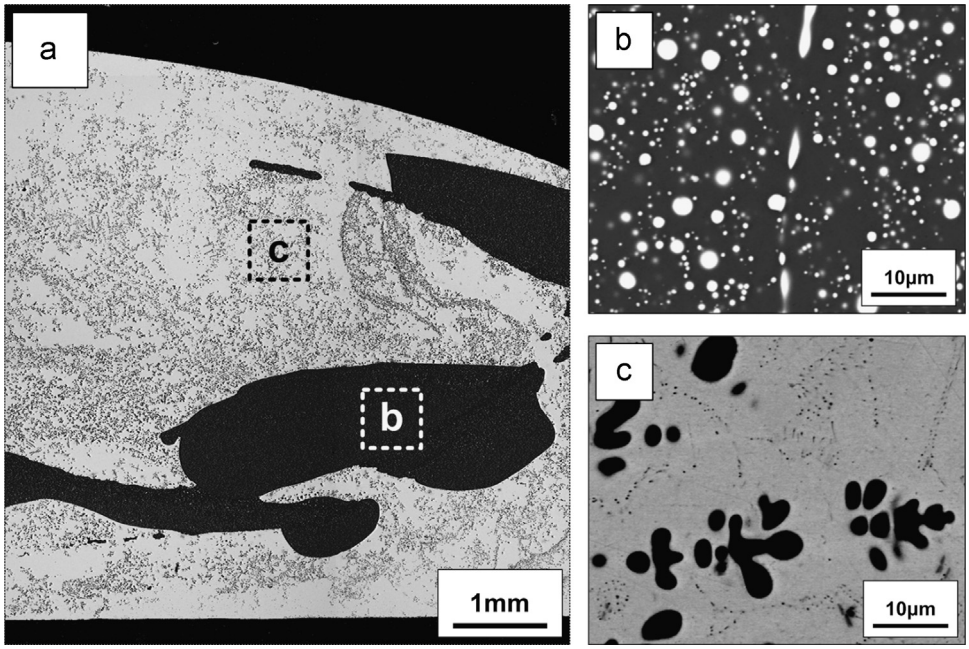

Fig. 10. SEM images of arc-melted $Nd_{40}Ti_{60}$. (a) Overview showing large-scale heterogeneities due to decomposition into Ti-rich (dark) and Nd-rich (grey) liquids. (b) Enlarged image of the Ti-rich liquid region showing a Nd-rich liquid (bright) and a Ti-phase (dark). (c) Enlarged image of the Nd-rich liquid showing Ti-dendrites (dark), Nd-rich eutectic regions and Nd-grains (grey).

Table 3
Chemical compositions of phases as determined by SEM/EDX analysis. The values for the different alloys represent an average over several sample points or areas in the case of the liquids L_1, L_2 and the eutectic regions e_1. The scatter in chemical compositions is taken as error bar.

Alloy	L_1 (at% Ti)	L_2 (at% Ti)	e_1 (at% Ti)	Nd (at% Ti)	Ti (at% Ti)
$Nd_{95}Ti_5$	–	–	1.3	0.4	–
$Nd_{80}Ti_{20}$	6.9	90.9	1.5	0.6	99.9
$Nd_{60}Ti_{40}$	6.8	91.1	1.5	0.5	99.8
$Nd_{40}Ti_{60}$	10.4	90.1	1.8	–	–
Nd_5Ti_{95}	–	89.2	–	–	99.7
Average	8.0 ± 3.0	90.3 ± 1.0	1.5 ± 0.5	0.5 ± 0.1	99.8 ± 0.1

the primary phase for the $Nd_{95}Ti_5$ alloy is in accordance with the calculated eutectic composition. About 0.2 at% Nd is found to be dissolved in the Ti(Nd) dendrites. On the other side about 0.5 at% Ti is dissolved in the Nd grains. The occurrence of the Nd-grains is probably due to metastable undercooling of the Nd(Ti) melt.

The experimental values as determined by SEM/EDX are summarized in Table 3 and also presented in Fig. 8.

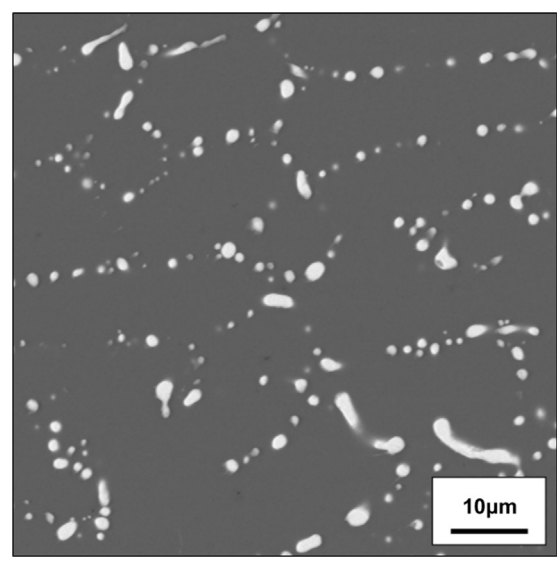

Fig. 11. SEM image of arc-melted Nd_5Ti_{95} alloy. Ti-grains (dark) and round shaped Nd phase (white) at the grain boundaries can be seen according to the primary formation of bcc-Ti from the melt.

4. Thermodynamic calculations

The thermodynamic assessment of the binary Nd–Ti system was done by the CALPHAD method [21]. The Gibbs free energy of each element is defined with respect to its stable magnetically disordered state at 298.15 K and 10^5 Pa (stable element reference). In the SGTE-Pure database 4.4 [22], the Gibbs free energy is represented as a power series in terms of the temperature in the form:

$$^0G(T) = AT\ln(T) + \sum_{n=-1} C_n T^n, \tag{2}$$

where A, and C_n are coefficients and n represents a set of integers. The reference states for pure neodymium and titanium are dhcp-Nd and hcp-Ti, respectively. It should be noted that there are no data for dhcp-Ti available in the SGTE pure database. The difference between hcp-Ti and dhcp-Ti was chosen as 1000 J/mole, which is approximately the same value as in the Nd system.

The liquid and the disordered solid solutions of the bcc, hcp and dhcp phases are described using a substitutional solution model, the Gibbs free energy being defined as

$$G^{\varphi} = \sum_{i=1}^{2} x_i {}^0G_i + G^{id} + G^{ex} \quad G^{id} = RT \sum_{i=1}^{2} x_i \ln(x_i)$$

$$G^{ex} = \sum_{i=1}^{2} \sum_{j>i}^{2} x_i x_j \sum_{v=0}^{n} {}^vL_{i,j}(x_i - x_j)^v, \tag{3}$$

where ^0G_i is the reference Gibbs free energy of the pure element i, i.e., Nd or Ti in the corresponding phase φ, G^{id} is the ideal mixing contribution, G^{ex} is the excess energy, x_i and x_j are the mole

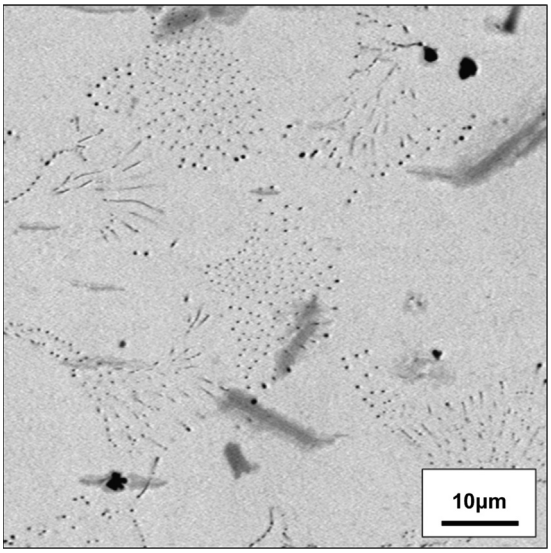

Fig. 12. SEM image of arc-melted $Nd_{95}Ti_5$ alloy. Primary Ti-dendrites (dark) and eutectic regions can be seen. Nd-grains (grey) occur probably due to metastable undercooling of the Nd(Ti) melt.

Table 4
Summary of thermodynamic parameters describing the Nd–Ti system referred to the stable element reference H^{SER} (T=298.15 K, P=100.000 kPa). Values are given in SI units (J, mol, and K).

Parameter	Ref.
$G^{hcp}_{Nd}, G^{dhcp}_{Nd}, G^{bcc}_{Nd}, G^{liquid}_{Nd}, G^{hcp}_{Ti}, G^{bcc}_{Ti}, G^{liquid}_{Ti}$	[22]
Liquid	
${}^{0}L^{liq}_{Nd,Ti} = +97687.38 - 29.51124T$	This work
${}^{1}L^{liq}_{Nd,Ti} = -1439.56$	This work
hcp	
${}^{0}L^{hcp}_{Nd,Ti} = +33238.18 + 20.15756T$	This work
${}^{1}L^{hcp}_{Nd,Ti} = +2415.88$	
bcc	
${}^{0}L^{bcc}_{Nd,Ti} = +60551.77 + 10.678T$	This work
${}^{1}L^{bcc}_{Nd,Ti} = -3891.40$	This work
dhcp	
$G^{dhcp}_{Ti} = G^{hcp}_{Ti} + 1000$	This work
${}^{0}L^{dhcp}_{Nd,Ti} = +73871.93$	

fractions of the constituents i, and j, respectively. ${}^{\nu}L_{i,j}$ is the binary interaction parameter of components i and j of an order ν. R is the gas constant, and T is the absolute temperature.

The thermodynamic model parameters for the binary system Nd–Ti were evaluated on the basis of the experimental information of this work using the least-squares optimization module Parrot of the software Thermo-Calc [23]. First, the parameters of *bcc* and liquid phase were optimized to fit temperatures and phase compositions in two reactions: $L_2 \leftrightarrow L_1 + bcc$-Ti and $L_1 \leftrightarrow bcc$-Nd$+bcc$-Ti. The parameters of *hcp* and *dhcp* were assessed to fit data for the reactions *bcc*-Nd$+$*bcc*-Ti$\leftrightarrow$*hcp*-Ti and *bcc*-Nd$\leftrightarrow$*dhcp*-Nd$+$*hcp*-Ti.

The final set of thermodynamic parameters was obtained by simultaneous optimisations of all mixing parameters. They are presented in Table 4. The computed Nd–Ti phase diagram using the present set of thermodynamic parameters is shown in Fig. 8 along with the experimental results. The calculated temperatures and phase compositions of the invariant equilibria are given in Table 5. The calculated coordinates of the miscibility gap critical point are 2120 K and 52.32 at%Ti.

The modelling results agree well with the experimental data obtained in the present work. It should be mentioned that the obtained diagram of the Nd–Ti system is quite similar to the phase diagram of the Gd–Ti system [8]. The assessed thermodynamic parameters are also similar for both systems.

Table 5
Invariant reactions of the Nd–Ti system.

Invariant reaction	T (K)		Calc. composition of phases (at% Ti)		
	Calc	Exp	1	2	3
$L_2 \leftrightarrow L_1 +$ bcc-Ti	1821	1840 ± 20	92.78	8.96	99.69
$L_1 \leftrightarrow$ bcc-Nd + bcc-Ti	1286	1278 ± 5	0.31	0.14	99.93
bcc-Nd + bcc-Ti ↔ hcp-Ti	1163	1160 ± 5	0.08	99.96	99.66
bcc-Nd ↔ dhcp-Nd + hcp-Ti	1127	1126 ± 5	0.07	0.03	99.69

5. Conclusions

Experimental and thermodynamic assessment of the binary Nd–Ti system has been carried out. Combining the conical nozzle levitation technique with in situ high-energy synchrotron X-ray diffraction enabled to directly observe the phase equilibria at elevated temperatures. The monotectic character of the phase diagram was proven. The coexistence of two liquids is directly observed by the occurrence of two diffuse diffraction maxima and experimental data of the binodal curve could be extracted. The experimental data confirm the general form as known from literature, however different values were obtained for the extension of the miscibility gap and transition temperatures. The experimental in situ XRD observations together with microstructure investigations of cast alloys and DSC led to an essentially improved new Nd–Ti phase diagram. Based on the experimental results, a self-consistent thermodynamic description could be developed.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2014.08.002.

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